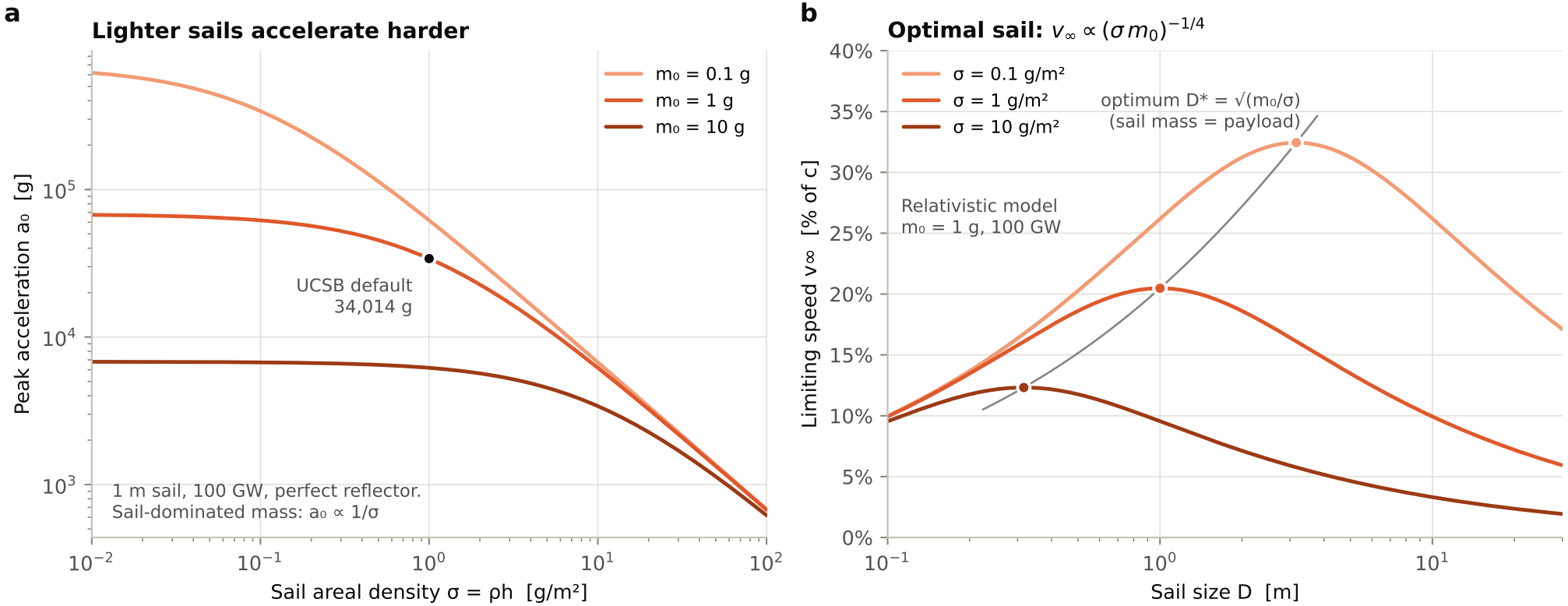
a
Lighter sails accelerate harder
m₀ = 0.1 g
m₀ = 1 g
m₀ = 10 g
Peak acceleration a₀ [g]
10⁵
10⁴
10³
UCSB default
34,014 g
1 m sail, 100 GW, perfect reflector.
Sail-dominated mass: a₀ ∝ 1/σ
10⁻²
10⁻¹
10⁰
10¹
10²
Sail areal density σ = ρh [g/m²]
b
Optimal sail: v∞ ∝ (σ m₀)^−1/4
σ = 0.1 g/m²
σ = 1 g/m²
σ = 10 g/m²
optimum D* = √(m₀/σ)
(sail mass = payload)
Relativistic model
m₀ = 1 g, 100 GW
Limiting speed v∞ [% of c]
40%
35%
30%
25%
20%
15%
10%
5%
0%
10⁻¹
10⁰
10¹
Sail size D [m]